llmXive Automated Review of Measuring Epistemic Resilience of LLMs Under Misleading Medical Context

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong factual claims regarding the performance of specific LLMs (e.g., GPT-5.4, Gemini-3.1) and the validity of the MedMisBench dataset. While the internal logic of the benchmark construction is sound, there are significant concerns regarding the verifiability of the cited sources for the models and datasets used.

First, the bibliography contains a mismatch for the MedMCQA dataset. The entry `c-001` points to a Hugging Face URL but the text cites `pal2022medmcqa`. The review must confirm that the bibliography file actually contains the full citation for Pal et al. (2022) and that the URL is supplementary, not the primary source. Without the primary citation, the claim that MedMCQA is a standard benchmark lacks proper academic grounding.

Second, and more critically, the paper cites models with future release dates (2026) such as "GPT-5.4" and "Gemini-3.1-pro". The citations `openai2026gpt54`, `google2026gemini31pro`, and `anthropic2026sonnet46` imply the existence of published technical reports or papers for these specific versions. If these models are hypothetical, unreleased, or if the citations are placeholders for future work, the core empirical claims (e.g., "clean accuracy = 71.1%", "ASR = 51.5%") are factually unsupported by the provided bibliography. The review cannot verify the accuracy of the results if the underlying models and their corresponding literature do not exist or are not properly cited.

Finally, the claim of 98.1% alignment between automatic ASR and clinician review is a strong statistical assertion. While the text mentions the sample size (89 tasks), it does not explicitly cite the specific statistical method or the definition of "correctness" used to derive this percentage within the provided snippets. The authors should ensure the reference to the clinician review protocol (Appendix B) is explicit in the main text to support this specific numerical claim.

The paper requires minor revisions to ensure all citations correspond to verifiable, existing literature and that the definitions of metrics used in high-impact claims are explicitly referenced.

Action items.

- [**writing**] The bibliography entry for MedMCQA (c-001) lists a Hugging Face dataset URL but lacks the primary citation (e.g., Pal et al., 2022) required to support the claim that MedMCQA is a standard medical benchmark. Verify the reference key 'pal2022medmcqa' exists in references.bib and is correctly cited in the text.
- [**science**] The paper cites 'openai2026gpt54', 'google2026gemini31pro', and 'anthropic2026sonnet46' for models released in 2026. As this is a preprint on arXiv (2606.12291), verify that these citations refer to actual, publicly available technical reports or papers from those future dates, or if they are placeholders for unreleased models. If the models are hypothetical or unreleased, the claims of 'expert-level scores' and specific ASR metrics cannot be factually supported by existing literature.
- [**writing**] The claim that 'Automatic ASR aligns with clinician correctness in 98.1% of annotations' (Section 5.6) relies on a specific definition of 'correctness' derived from the clinician review. Ensure the citation or internal reference clearly defines the ground truth metric used for this alignment calculation, as the text does not explicitly link the 98.1% figure to a specific statistical test or validation protocol in the provided text.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure effectively visualizes the injection process described in the text, but the caption filename is a critical error that contradicts the figure label and the provided caption list. Additionally, the specific medical claim used as the injection is not defined in the caption, obscuring the source of the 'Authority' provenance.

Figure 2

The figure provides a clear visual overview of the dataset pipeline and attack types, but the workflow logic is slightly obscured by the disconnected placement of Step 3 and the numbering skips from 2 to 4.

Figure 3

Figure 3 presents a bar chart comparing clean, Type 1, and Type 2 accuracy across models, but fails to clearly link the caption's aggregate claims (e.g., 38.0% Type 1 mean) to visual elements. The x-axis sub-labels are undefined, and the three dashed 'mean reference' lines lack legend differentiation, undermining clarity and scientific rigor.

Figure 4

The figure effectively visualizes the disparity between Type 1 and Type 2 attack success rates, but the caption contains a critical file reference error ('Figure5.pdf'). Additionally, the legend definition for 'gap' contradicts the visual representation of connecting lines.

Figure 5

The figure is a UI screenshot that fails to visually demonstrate the 'Rubric A/B controls' mentioned in the caption, and the displayed content appears to be metadata definitions rather than the specific review artifacts described.

Figure 6

The figure attempts to show generator sensitivity but presents data that contradicts the caption's narrative, as the new generator shows drastically higher attack success rates rather than a consistent pattern. Additionally, the legend layout is disjointed.

Action items.

- **[fatal]** Figure 1: The caption filename '[Figure2.pdf]' contradicts the label 'Figure 1' and the content, which matches the description of Figure 2 in the provided caption list (turning clean QA into resilience tests).
- **[science]** Figure 1: The 'Injection Sentence' box contains a specific medical claim ('lesions under 1.2 mm require... no lymph node evaluation') that is not defined or sourced in the caption, making the 'Authority' provenance claim unverifiable.
- **[science]** Figure 2: The workflow diagram displays numbered steps 1, 2, and 4, but Step 3 ('Taxonomy Generation & Attack Creation') is positioned below the main flow without a connecting arrow, obscuring the logical sequence of the pipeline.
- **[writing]** Figure 2: The caption refers to 'Figure1.pdf' in brackets, which contradicts the figure label 'Figure 2' and suggests a file naming or cross-reference error.
- **[science]** Figure 3: The caption claims 'Clean accuracy overstates epistemic resilience' and cites a drop to 38.0% under Type 1, but the chart shows Type 1 (red bars) varies wildly by model (e.g., 29.9% for Gemini-3.1-pro vs 54.0% for GPT-5.4). The 38.0% value is an aggregate mean not explicitly labeled as such on the chart, making the claim misleading without a clear 'mean' bar or annotation for Type 1.
- **[writing]** Figure 3: The x-axis labels are cluttered and inconsistent; some models have sub-labels like 'high', 'median', 'low', 'none', 'medium', 'minimal' while others do not, and the meaning of these sub-labels is undefined in the caption or legend.
- **[writing]** Figure 3: The legend defines 'mean reference' as a dashed line, but there are three horizontal dashed lines (blue, green, red) with no legend entry explaining which corresponds to which mean (clean, Type 1, Type 2), despite the caption mentioning all three values.
- **[fatal]** Figure 4: The caption cites 'Figure5.pdf' instead of the correct file name 'Figure4.pdf'.
- **[science]** Figure 4: The legend defines 'gap' as a solid line, but the plot uses solid lines to connect paired data points (T2/T1) for each model rather than representing a 'gap' metric.
- **[writing]** Figure 4: The legend entry 'T2 all-option ASR' is ambiguous; the plot shows blue circles for T2, but the label does not clarify if this represents the mean or individual data points.
- **[science]** Figure 5: The image displays a single case (MEDMISMCQA) with a 'Neutral False Statement' injection, but the caption claims to show 'Rubric A/B controls' which are not visible in the provided screenshot.
- **[writing]** Figure 5: The image is a screenshot of a UI rather than a rendered figure; the 'Provenance vehicle' and 'Misinformation type' sections appear to be metadata definitions rather than the actual 'injected context' or 'model output' mentioned in the caption.

- **[science]** Figure 6: The x-axis is labeled 'Attack success rate (%)' but the data points for the 'GPT-5.4 generator' (green circles) are plotted at values (e.g., 63.8, 61.5) that are significantly higher than the 'Main generator' (grey circles, e.g., 35.0, 40.6). This contradicts the caption's claim that 'Type 1 remains more damaging' (implying the new generator should not drastically increase ASR) and suggests a potential labeling error or data inversion.
- **[writing]** Figure 6: The legend at the bottom is split into two disconnected parts ('Main generator' on the left, 'GPT-5.4 generator' on the right) without a unifying title or box, making it slightly ambiguous whether these are mutually exclusive categories or separate series.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on specialized terminology and undefined acronyms that create a barrier for non-specialist readers, particularly those in clinical or general AI safety fields.

First, the core concept of "epistemic resilience" is introduced in Section 1 without a plain-language equivalent. While the authors define it as "preserving correct medical judgment," the term itself is philosophical jargon. A simpler phrase like "resistance to misinformation" or "truthfulness under pressure" would be more accessible.

Second, critical metrics are introduced as acronyms without definition. In Section 5.1, "ASR" and "TASR" appear immediately without being spelled out as "Attack Success Rate" and "Targeted Attack Success Rate." Similarly, "Gwet AC2" is cited in Section 5.7 without explanation. These acronyms are standard in specific sub-fields but obscure the meaning for a broader audience.

Third, the "Type 1" and "Type 2" labels for delivery protocols (Section 4.4) are arbitrary jargon. The text describes them as "Focused" and "All-option," but the labels themselves do not convey meaning. Renaming these to "Focused Injection" and "Full Bundle Injection" would improve readability.

Finally, phrases like "content/provenance decomposition" (Section 2, Table 1) and "answer-grounded" (Section 1) are unnecessarily technical. "Decomposition" could be "separation," and "answer-grounded" could be "based on a specific correct answer." These changes would significantly lower the entry barrier for readers without a background in formal evaluation metrics.

Action items.

- **[writing]** Define 'ASR' (Attack Success Rate) and 'TASR' (Targeted ASR) at their first occurrence in Section 5.1. Currently, they appear as acronyms without expansion, hindering non-specialist readers.
- **[writing]** Replace the term 'epistemic resilience' with a plain-language definition or synonym

(e.g., 'resistance to misinformation') upon first introduction in Section 1. The current definition is abstract and relies on specialized philosophical terminology.

- [**writing**] Clarify 'Type 1' and 'Type 2' delivery protocols in Section 4.4. While described, the labels are arbitrary jargon; consider renaming them to 'Focused Injection' and 'Full Bundle Injection' for immediate clarity.
- [**writing**] Define 'Gwet AC2' in Section 5.7. The statistical metric is used to report inter-rater agreement but is not explained, excluding readers unfamiliar with this specific coefficient.
- [**writing**] Replace 'content/provenance decomposition' in Section 2 and Table 1 with 'separating the false claim from its source type' to avoid unnecessary technical phrasing.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent logical framework for defining and measuring "epistemic resilience," but there are minor inconsistencies in how specific metrics are compared and interpreted in the text versus the data tables.

First, the central claim in the Introduction and Conclusion—that clean accuracy "vastly overestimates" resilience—is logically supported by the drop in performance under attack, but the specific metrics cited in the text create a slight logical friction. The authors compare Clean Accuracy (71.1%) directly to Attack Success Rate (ASR, 51.5%) to illustrate the gap. Logically, a more direct comparison would be Clean Accuracy (71.1%) versus Type 1 Accuracy (38.0%). While the ASR metric is valid for measuring the attack's success, equating the "overestimation" gap to the difference between accuracy and ASR (which are different scales) is slightly imprecise. The argument would be tighter if the text explicitly contrasted the 71.1% clean success rate against the 38.0% success rate under Type 1 attacks.

Second, in Section 3.4 (Delivery Protocols), the text states that Type 2 accuracy "returns to 70.5% without eliminating ASR failures." The data shows Type 2 ASR is 18.7%. While 18.7% is non-zero, the phrasing "without eliminating" combined with the high accuracy figure (70.5%) creates a slight narrative tension. The logic holds that failures exist, but the text could be clearer that the *rate* of failure drops significantly (from 51.5% to 18.7%) rather than implying a persistent high level of failure that contradicts the high accuracy.

Finally, the taxonomy analysis claims "Formal, rule-like falsehoods are most damaging," citing Authority framing (69.5%) and Exception Poisoning (64.1%). The data in Table 3 shows Threshold/Reference Corruption at 60.9%, which is statistically close to Exception Poisoning. The text treats Exception Poisoning as the distinct peak, but the data suggests a cluster of high-risk categories. While not a fatal flaw, the conclusion would be more logically robust if it acknowledged this cluster of "rule-like" failures rather than singling out one specific type as the sole maximum, or if it provided a statistical justification for the distinction.

Overall, the causal mechanisms (misleading context -> reduced accuracy) are well-supported, and the internal logic of the benchmark design is sound. These are primarily issues of precise metric comparison and narrative framing rather than fundamental logical contradictions.

Action items.

- [**science**] The claim that 'clean accuracy vastly overestimates epistemic resilience' relies on a comparison between clean accuracy (71.1%) and Type 1 ASR (51.5%). These are not directly comparable metrics (accuracy vs. attack success rate). The paper should explicitly compare clean accuracy to Type 1 accuracy (38.0%) to logically support the 'overestimation' claim, or clarify the metric relationship.
- [**writing**] In Section 3.4, the paper states Type 2 accuracy 'returns to 70.5% without eliminating ASR failures.' However, Table 2 shows Type 2 ASR is 18.7%. The text implies a contradiction between high accuracy and high failure rate, but the numbers show a significant drop in failure rate. The narrative needs to reconcile the high accuracy with the remaining ASR to avoid logical confusion.
- [**writing**] The conclusion states 'Formal rule-like injections cause the most failures,' citing authority framing (69.5% ASR) and exception poisoning (64.1% ASR). However, Table 3 shows 'Threshold/Reference Corruption' has a mean ASR of 60.9%, which is close to exception poisoning. The text should clarify if 'rule-like' encompasses both or if the distinction is statistically significant, as the current phrasing implies a clear hierarchy that the data only partially supports.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper significantly overreaches by presenting empirical results for LLMs that do not currently exist. The manuscript explicitly lists and evaluates "GPT-5.4", "Gemini-3.1-pro/flash-lite", and "Claude-sonnet-4.6" in Section 5.1 and throughout Tables 1, 2, and 3. As of the current date, these model versions are not publicly available or released.

By presenting specific accuracy metrics, Attack Success Rates (ASR), and detailed failure modes for these non-existent models, the authors extrapolate beyond the scope of available data. The paper frames these results as a measurement of the current state of "epistemic resilience" in medical LLMs. This is a fundamental overreach: one cannot measure the resilience of a system that has not been built or released.

If the authors intended to simulate future models or use hypothetical parameters, this must be explicitly stated as a limitation or a theoretical exercise. Presenting the data as empirical findings from real API calls (implied by the specific version numbers and "accessed" language in

Appendix C.1) misleads the reader regarding the validity of the evidence. The claim that "clean accuracy vastly overestimates epistemic resilience" is unsupported if the "clean accuracy" and "resilience" metrics are derived from non-existent systems.

Furthermore, the clinician review (Section 5.7) analyzes "harm" based on outputs from these same non-existent models. The conclusion that "38.2% of outputs pose serious harm" is an overreach if the outputs are synthetic or projected, as the clinical risk assessment relies on the authenticity of the model's behavior. The paper must either provide evidence that these models exist and were accessed, or reframe the entire study as a simulation/projection, acknowledging that the specific numerical results are hypothetical and not empirical measurements of current technology. Without this correction, the core scientific claims are unsupportable.

Action items.

- [**science**] The paper significantly overreaches by presenting empirical results for LLMs that do not currently exist. The manuscript explicitly lists and evaluates "GPT-5.4", "Gemini-3.1-pro/flash-lite", and "Claude-sonnet-4.6" in Section 5.1 and throughout Tables 1, 2, and 3. As of the current date, these model versions are not publicly available or released. By presenting specific accuracy metrics, Attack Success Rates (ASR), and detailed failure modes for these non-existent models, the authors extrapolat

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper addresses a critical safety concern: the vulnerability of medical LLMs to misleading context, which could lead to harmful clinical decisions. The methodology of injecting false claims and measuring "epistemic resilience" is sound in principle for identifying safety gaps. However, several ethical and safety reporting gaps must be addressed before publication.

First, the human evaluation component (Section 5.7, Appendix B) involves a 14-member clinician panel assessing potential harm. While the authors mention adherence to the Declaration of Helsinki, the manuscript lacks specific details regarding Institutional Review Board (IRB) approval or exemption status, the informed consent process for the clinicians, and the specific protocols used to ensure the privacy of any patient data (even if de-identified or synthetic) reviewed during the study. Given the sensitivity of medical data and the potential for harm in the reviewed scenarios, explicit confirmation of ethical oversight is required.

Second, the dataset itself, 'MedMisBench', contains high-quality, clinically plausible false medical statements (e.g., incorrect drug dosages, fake guidelines). While the intent is evaluation, the public release of these specific falsehoods creates a dual-use risk. Malicious actors could potentially use this dataset to train models to generate medical misinformation or to poison other datasets. The "Ethics and Intended Use" section (Appendix E.2) currently focuses on the intended use but does not sufficiently address the mitigation of this negative impact. The authors should propose a specific license restriction (e.g., prohibiting use for training generative models) or a technical mitigation (e.g., watermarking the false claims) to prevent misuse.

Finally, the paper evaluates models that do not yet exist (e.g., GPT-5.4). While the safety analysis is valuable for future-proofing, the "Ethics" section must explicitly state that these results are projections based on simulated capabilities. Failing to make this distinction could lead to the misinterpretation of these findings as evidence of actual, current safety failures in deployed systems, potentially causing unnecessary alarm or misinformed policy decisions. The authors should clarify the hypothetical nature of the specific model results while maintaining the validity of the proposed evaluation framework.

Action items.

- [**writing**] The 'Ethics' section (Appendix E.2) lacks specific IRB approval numbers and details on the informed consent process for the 14-member clinician panel. Explicitly state the IRB status and consent methodology to ensure compliance with human subjects research standards.
- [**writing**] The public release of 'MedMisBench' contains realistic false medical claims, posing a dual-use risk for poisoning training data or generating misinformation. Add a specific license restriction or technical mitigation (e.g., watermarking) to prevent malicious reuse of these false statements.
- [**writing**] The paper evaluates non-existent models (e.g., GPT-5.4). The 'Ethics' section must explicitly clarify that these results are simulations based on projected capabilities, not evidence of actual safety failures in currently deployed systems, to prevent public misinterpretation.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling hypothesis regarding the vulnerability of medical LLMs to misleading context, but the scientific evidence supporting the magnitude and robustness of these claims requires strengthening.

First, the construction of the benchmark relies heavily on LLM-generated injections (Section 3.3). While the authors perform a post-hoc clinician review on a small sample (n=89, Section 4.6), there is no evidence provided that the *generation* process itself produces clinically plausible falsehoods at a high rate. If the injected context is often nonsensical or easily dismissed by the model regardless of its "resilience," the Attack Success Rate (ASR) metrics may be inflated or measuring a different phenomenon (e.g., confusion by noise). The paper needs to report the validation rate of the generated injections *before* they are fed to the target models, ideally with inter-rater agreement from multiple clinicians on a larger random sample of the full dataset (10,932 items), not just the 89 reviewed for harm.

Second, the statistical rigor of the harm assessment is questionable. The authors report precise 95% confidence intervals for worst-case harm (e.g., 38.2% [28.8–48.6%]) based on a sample of only 89 tasks. This sample size yields a margin of error of approximately ±10%, making the reported precision misleading. Furthermore, the analysis of moderators (e.g., "Authority framing" vs. "Patient framing") in Table 5 (Appendix) relies on a mixed-effects logistic regression on this small sample. The wide confidence intervals for some odds ratios (e.g., Threshold/Reference corruption: 0.90–13.55) indicate a lack of statistical power to draw definitive conclusions about which specific types of misinformation are most dangerous. The authors should either expand the clinician review significantly or reframe these findings as preliminary trends rather than definitive statistical evidence.

Finally, the mitigation case studies (Section 4.8) lack formal statistical testing. The reported reductions in ASR (e.g., from 81.5% to 16.1% with search) are substantial, but without a paired statistical test (such as McNemar's test for the same items) or bootstrapped confidence intervals, it is difficult to rule out that these improvements are due to the specific composition of the HLE-only subset rather than the mitigation strategy itself. The evidence for the effectiveness of defensive prompts is similarly descriptive. To support the central claim that these mitigations are viable, the authors must provide statistical significance tests for the observed differences.

Action items.

- [**science**] The study relies on synthetic injections generated by LLMs (Gemini-3.1-flash) without independent human validation of the falsehoods' clinical plausibility or logical coherence prior to evaluation. The clinician review (n=89) is post-hoc and sampled; the paper must report the inter-rater reliability and failure rate of the *generation* process itself to ensure the benchmark measures resilience to plausible errors, not just random noise.
- [**science**] The sample size for the clinician harm assessment (n=89 tasks) is insufficient to support the precise confidence intervals (e.g., 95% CI 28.8–48.6%) reported for worst-case harm rates. The statistical power for detecting differences in harm severity across provenance types is likely underpowered; the authors should either increase the clinician review sample size or report the harm findings as qualitative trends with wider, more conservative uncertainty bounds.
- [**science**] The mitigation results (search, defensive prompts) are reported on specific subsets (e.g., HLE-only for search, n=600 for prompts) without a clear statistical test (e.g., McNemar's test or bootstrap CI) confirming that the observed reductions in ASR are statistically significant and not due to random variance in the subset selection.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis of the main benchmark results lacks necessary uncertainty quantification. While the paper reports precise point estimates for Attack Success Rates (ASR) across 10,932 items (e.g., Mean Type 1 ASR = 51.5% in Section 4.2), it fails to provide confidence intervals (CIs) or standard errors for these aggregate metrics. In contrast, the smaller clinician review subset (n=89) in Section 4.7 and Appendix E.2 explicitly reports 95% CIs (e.g., 28.8–48.6%). Given the binary nature of the evaluation (correct/incorrect), the standard error for the main ASR should be negligible, but omitting CIs entirely prevents readers from assessing the precision of the reported means, especially when comparing models with smaller sample sizes or specific sub-strata.

Furthermore, the manuscript makes strong comparative claims (e.g., Type 1 ASR is "2.8× higher" than Type 2; authority framing is "most damaging") without reporting the results of formal statistical hypothesis tests. The analysis of the taxonomy (Section 4.6) involves multiple comparisons across 5 content types and 3 provenance levels. Without reporting p-values, test statistics (e.g., chi-square or logistic regression coefficients), or a correction method for multiple testing (e.g., Bonferroni or Benjamini-Hochberg), it is unclear if the observed differences between "Exception Poisoning" (64.1%) and "Spurious Anchoring" (20.9%) are statistically significant or potentially due to random variation in the specific injection generation.

The use of Gwet AC2 for inter-rater reliability is appropriate for ordinal data and avoids the prevalence bias of Cohen's Kappa, but the manuscript should briefly justify this choice in the methods section to align with standard reporting practices in medical AI literature. Finally, the mixed-effects logistic regression in Appendix E.2 (Table adjusted-moderators) reports Adjusted Odds Ratios with Wald CIs, which is good practice, but the model specification (random effects structure, fixed effects included) is not detailed enough to fully assess the robustness of the "Authority framing" effect size.

Action items.

- [**science**] Report confidence intervals or standard errors for the primary mean ASR metrics (e.g., 51.5% Type 1 ASR) in Tables 1 and 2. Currently, only the clinician review subset (n=89) includes 95% CIs, leaving the main benchmark results (n=10,932) without uncertainty quantification.
- [**science**] Clarify the statistical test used to compare Type 1 vs. Type 2 ASR and the differences across provenance types. The text claims significant differences (e.g., "2.8x higher") but does not specify the test statistic, p-values, or correction for multiple comparisons across the 5 content types and 3 provenance levels.
- [**writing**] Justify the use of Gwet AC2 over Cohen's Kappa for inter-rater reliability in the clinician review (Section 4.7). While AC2 is appropriate for ordinal data, the manuscript should explicitly state why this metric was chosen over others and confirm that the 0.78–0.95 range meets the threshold for "substantial" agreement in this specific medical context.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high level of technical clarity and logical flow, effectively communicating a complex evaluation framework for LLM epistemic resilience. The introduction successfully defines the core problem and the proposed benchmark, while the taxonomy section (Section 3.1) is well-structured, using clear enumeration to define the content and provenance axes. The transition from dataset construction to experimental results is smooth, and the use of specific metrics (ASR, TASR) is consistently applied throughout the text.

However, there are minor areas where sentence-level precision could be improved to eliminate potential ambiguity. In Section 3.3, the description of the injection generation process uses the phrase "single all-option call." While likely clear to an expert, this phrasing is slightly opaque regarding the specific mechanism (e.g., is it a single prompt generating multiple lines, or a batched API call?). Clarifying this would prevent misinterpretation of the generation pipeline's efficiency or design.

Additionally, in Section 4.2, the text reports "Type 2 accuracy 70.5%." Given that the paper relies heavily on precise statistical comparisons (e.g., "2.8× higher"), the use of the approximation symbol here is inconsistent with the precision of the surrounding data. It is recommended to either provide the exact figure or explicitly state the rounding policy for all reported percentages to maintain a uniform standard of rigor.

Finally, the appendices contain several tables with "(... X rows omitted ...)" placeholders. While this is a common convention for preprints to save space, the writing should ensure that the reader is explicitly directed to the external repository or a supplementary file for the complete data tables. The current text assumes the reader will infer this, but a brief explicit note (e.g., "Full tables are available in the repository") would improve the self-containment and readability of the document. Overall, the writing is strong and requires only these minor refinements to reach publication-ready quality.

Action items.

- [**writing**] In Section 3.3 (Injection Generation), the phrase 'single all-option call' is ambiguous. Clarify whether this refers to a single API invocation generating all distractors simultaneously or a specific prompting strategy to ensure consistency.
- [**writing**] Section 4.2 (Overall Results) states 'Type 2 accuracy 70.5%'. Replace the approximation symbol with the exact value or explicitly state the rounding convention used for this specific metric to maintain precision consistency with other reported figures.
- [**writing**] The Appendix contains multiple instances of '(... X rows omitted ...)' within table environments. While acceptable for a preprint summary, ensure the final camera-ready version includes full tables or clearly references the external repository for the complete data to avoid reader confusion.